Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (1): Design

Generation Category (1): Design

Load Diversity Factor: None

	Swing	V-Control	Load	Total
Number of Buses:	1	0	22	23

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	7	0	0	15	0	0	22

Method of Solution:	Adaptive Newton-Raphson Method
Maximum No. of Iteration:	99
Precision of Solution:	0.0001000
System Frequency:	50.00 Hz
Unit System:	Metric
Project Filename:	data_center_study_bootcamp
Output Filename:	D:\Software\ETAP\data_center_study_bootcamp\Untitled.lfr

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
20 kV MVP-L1-CSR-A	20.000	1	100.0	0.0								
Bus PLN-A	20.000	1	100.0	0.0								
Bus2	0.400	1	100.0	0.0	0.760	0.250	0.190	0.062				
Bus3	20.000	1	100.0	0.0								
Bus4	0.400	1	100.0	0.0								
Bus5	20.000	1	100.0	0.0								
Bus6	0.400	1	100.0	0.0								
Bus7	0.400	1	100.0	0.0	0.760	0.250	0.190	0.062				
Bus8	0.400	1	100.0	0.0								
Bus9	0.400	1	100.0	0.0	0.289	0.095	0.072	0.024				
Bus10	0.400	1	100.0	0.0								
Bus11	11.000	1	100.0	0.0								
Bus12	0.400	1	100.0	0.0	0.365	0.120	0.091	0.030				
Bus13	0.400	1	100.0	0.0								
Bus15	20.000	1	100.0	0.0								
Bus16	0.400	1	100.0	0.0	3.800	1.249	0.950	0.312				
Bus17	11.000	1	100.0	0.0								
Bus19	11.000	1	100.0	0.0								
Bus20	11.000	1	100.0	0.0								
LDMP-L2-C1	0.400	1	100.0	0.0	0.585	0.192	0.146	0.048				
MV-GSB-01	11.000	1	100.0	0.0								
MVSB-L2-CH1	11.000	1	100.0	0.0								
MVSB-L4-A1	11.000	1	100.0	0.0								
Total Number of Buses: 23					6.559	2.156	1.640	0.539	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus PLN-A	20.000	Swing	1	100.0	0.0					
						0.000	0.000			

Line/Cable/Busway Input Data

Line/Cable/Busway			ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)						
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable1	20NCUS3	300	83.0	0.0	2	75	0.076207	0.092700	
Cable2	20NCUS3	120	20.0	0.0	1	75	0.186940	0.106000	
Cable3	1,0NCUS3	300	100.0	0.0	8	75	0.076207	0.069700	
Cable4	20NCUS3	120	20.0	0.0	1	75	0.186940	0.106000	
Cable5	1,0NCUS3	300	100.0	0.0	8	75	0.076207	0.069700	
Cable6	11MCUS3	240	72.7	0.0	1	75	0.093470	0.090000	0.0001495
Cable7	1,0MCUN1	300	31.0	0.0	4	75	0.074200	0.112700	
Cable8	11MCUS3	240	72.2	0.0	1	75	0.093470	0.090000	0.0001495
Cable9	1,0MCUN1	300	20.0	0.0	13	75	0.074200	0.112700	
Cable10	11MCUS3	240	68.5	0.0	1	75	0.093470	0.090000	0.0001495
Cable11	20NCUS3	300	61.6	0.0	1	75	0.076207	0.092700	
Cable12	11MCUS3	300	35.3	0.0	1	75	0.076302	0.087400	0.0001646
Cable13	0,6MCUN1	300	34.0	0.0	13	75	0.147638	0.141076	
Cable14	11MCUS1	120	24.6	0.0	1	75	0.186940	0.118000	0.0001109
Cable15	20NCUS3	95	20.0	0.0	1	75	0.235582	0.109000	

Line / Cable / Busway resistances are listed at the specified temperatures.

2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
T-Loadbank01	3-Phase	5.500	11.000	0.400	7.15	8.50	0	0	0	0	0	7.1500	Dyn	0.000
TRF-HV-L1-A1	3-Phase	10.000	20.000	11.000	12.00	6.00	0	0	0	-2.500	0	12.0000	Dyn	0.000
TRF-L2-CH1	3-Phase	4.200	11.000	0.400	7.50	8.50	0	0	0	0	0	7.5000	Dyn	0.000
TRF-L8-A1	3-Phase	4.000	11.000	0.400	7.50	8.50	0	0	0	0	0	7.5000	Dyn	0.000
TRF-L8-A3	3-Phase	1.400	11.000	0.400	7.50	6.00	0	0	0	0	0	7.5000	Dyn	0.000
TRF-L10-A1	3-Phase	2.800	20.000	0.400	7.50	6.00	0	0	0	0	0	7.5000	Dyn	0.000
TRF-L10-A3	3-Phase	2.000	20.000	0.400	7.50	6.00	0	0	0	0	0	7.5000	Dyn	0.000

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
T-Loadbank01	2W XFMR	MV-GSB-01	Bus16	15.19	129.11	130.00	
TRF-HV-L1-A1	2W XFMR	Bus15	Bus17	19.23	115.41	117.00	
TRF-L2-CH1	2W XFMR	Bus11	Bus8	20.86	177.35	178.57	
TRF-L8-A1	2W XFMR	Bus19	Bus13	21.91	186.22	187.50	
TRF-L8-A3	2W XFMR	Bus20	Bus10	88.07	528.43	535.71	
TRF-L10-A1	2W XFMR	Bus5	Bus6	44.04	264.21	267.86	
TRF-L10-A3	2W XFMR	Bus3	Bus4	61.65	369.90	375.00	
Cable1	Cable	Bus PLN-A	20 kV MVP-L1-CSR-A	0.08	0.10	0.12	
Cable2	Cable	20 kV MVP-L1-CSR-A	Bus3	0.09	0.05	0.11	
Cable3	Cable	Bus4	Bus2	59.54	54.45	80.68	
Cable4	Cable	20 kV MVP-L1-CSR-A	Bus5	0.09	0.05	0.11	
Cable5	Cable	Bus6	Bus7	59.54	54.45	80.68	
Cable6	Cable	MVSB-L4-A1	Bus20	0.56	0.54	0.78	0.0013151
Cable7	Cable	Bus10	Bus9	35.94	54.59	65.36	
Cable8	Cable	MVSB-L4-A1	Bus19	0.56	0.54	0.77	0.0013057
Cable9	Cable	Bus13	Bus12	7.13	10.84	12.97	
Cable10	Cable	MVSB-L4-A1	MVSB-L2-CH1	0.53	0.51	0.73	0.0012391
Cable11	Cable	20 kV MVP-L1-CSR-A	Bus15	0.12	0.14	0.18	
Cable12	Cable	Bus17	MVSB-L4-A1	0.22	0.26	0.34	0.0007037
Cable13	Cable	Bus8	LDMP-L2-C1	24.13	23.06	33.38	
Cable14	Cable	MVSB-L2-CH1	Bus11	0.38	0.24	0.45	0.0003301
Cable15	Cable	MV-GSB-01	MVSB-L4-A1	0.39	0.18	0.43	

LOAD_FLOW_REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
20 kV MVP-L1-CSR-A	20.000	99.990	0.0	0.000	0.000	0.000	0.000	Bus PLN-A	-8.237	-3.646	260.1	91.4	
								Bus3	0.953	0.353	29.3	93.8	
								Bus5	0.953	0.342	29.2	94.1	
								Bus15	6.331	2.951	201.7	90.6	
* Bus PLN-A	20.000	100.000	0.0	8.238	3.647	0.000	0.000	20 kV MVP-L1-CSR-A	8.238	3.647	260.1	91.4	
Bus2	0.400	97.404	-2.1	0.000	0.000	0.940	0.309	Bus4	-0.940	-0.309	1466.7	95.0	
Bus3	20.000	99.989	0.0	0.000	0.000	0.000	0.000	20 kV MVP-L1-CSR-A	-0.953	-0.353	29.3	93.8	
								Bus4	0.953	0.353	29.3	93.8	
Bus4	0.400	98.152	-1.9	0.000	0.000	0.000	0.000	Bus2	0.946	0.315	1466.7	94.9	
								Bus3	-0.946	-0.315	1466.7	94.9	
Bus5	20.000	99.989	0.0	0.000	0.000	0.000	0.000	20 kV MVP-L1-CSR-A	-0.953	-0.342	29.2	94.1	
								Bus6	0.953	0.342	29.2	94.1	
Bus6	0.400	98.693	-1.4	0.000	0.000	0.000	0.000	Bus7	0.948	0.315	1461.6	94.9	
								Bus5	-0.948	-0.315	1461.6	94.9	
Bus7	0.400	97.947	-1.6	0.000	0.000	0.942	0.310	Bus6	-0.942	-0.310	1461.6	95.0	
Bus8	0.400	97.523	-4.7	0.000	0.000	0.000	0.000	LDMP-L2-C1	0.725	0.239	1130.2	95.0	
								Bus11	-0.725	-0.239	1130.2	95.0	
Bus9	0.400	96.945	-5.1	0.000	0.000	0.357	0.117	Bus10	-0.357	-0.117	559.0	95.0	
Bus10	0.400	97.144	-5.0	0.000	0.000	0.000	0.000	Bus9	0.357	0.118	559.0	94.9	
								Bus20	-0.357	-0.118	559.0	94.9	
Bus11	11.000	98.121	-4.0	0.000	0.000	0.000	0.000	MVSB-L2-CH1	-0.726	-0.250	41.1	94.6	
								Bus8	0.726	0.250	41.1	94.6	
Bus12	0.400	97.689	-4.5	0.000	0.000	0.452	0.149	Bus13	-0.452	-0.149	702.7	95.0	
Bus13	0.400	97.738	-4.4	0.000	0.000	0.000	0.000	Bus12	0.452	0.149	702.7	95.0	
								Bus19	-0.452	-0.149	702.7	95.0	
Bus15	20.000	99.978	0.0	0.000	0.000	0.000	0.000	20 kV MVP-L1-CSR-A	-6.331	-2.950	201.7	90.6	
								Bus17	6.331	2.950	201.7	90.6	-2.500
Bus16	0.400	95.096	-7.5	0.000	0.000	4.659	1.531	MV-GSB-01	-4.659	-1.531	7443.8	95.0	
Bus17	11.000	98.150	-3.9	0.000	0.000	0.000	0.000	MVSB-L4-A1	6.239	2.401	357.5	93.3	
								Bus15	-6.239	-2.401	357.5	93.3	
Bus19	11.000	98.126	-4.0	0.000	0.000	0.000	0.000	MVSB-L4-A1	-0.453	-0.153	25.6	94.7	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus13	0.453	0.153	25.6	94.7	
Bus20	11.000	98.127	-4.0	0.000	0.000	0.000	0.000	MVSB-L4-A1	-0.359	-0.126	20.3	94.3	
								Bus10	0.359	0.126	20.3	94.3	
LDMP-L2-C1	0.400	97.287	-4.8	0.000	0.000	0.724	0.238	Bus8	-0.724	-0.238	1130.2	95.0	
MV-GSB-01	11.000	98.108	-4.0	0.000	0.000	0.000	0.000	MVSB-L4-A1	-4.700	-1.875	270.7	92.9	
								Bus16	4.700	1.875	270.7	92.9	
MVSB-L2-CH1	11.000	98.125	-4.0	0.000	0.000	0.000	0.000	MVSB-L4-A1	-0.726	-0.250	41.1	94.6	
								Bus11	0.726	0.250	41.1	94.6	
MVSB-L4-A1	11.000	98.130	-4.0	0.000	0.000	0.000	0.000	Bus20	0.359	0.125	20.3	94.4	
								Bus19	0.453	0.152	25.5	94.8	
								MVSB-L2-CH1	0.726	0.249	41.1	94.6	
								Bus17	-6.238	-2.401	357.5	93.3	
								MV-GSB-01	4.701	1.875	270.7	92.9	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
20 kV MVP-L1-CSR-A	20.000										9.008	91.4	260.1	
Bus PLN-A	20.000										9.009	91.4	260.1	
Bus2	0.400		0.760	0.250	0.180	0.059					0.990	95.0	1466.7	
Bus3	20.000										1.016	93.8	29.3	
Bus4	0.400										0.997	94.9	1466.7	
Bus5	20.000										1.013	94.1	29.2	
Bus6	0.400										0.999	94.9	1461.6	
Bus7	0.400		0.760	0.250	0.182	0.060					0.992	95.0	1461.6	
Bus8	0.400										0.764	95.0	1130.2	
Bus9	0.400		0.289	0.095	0.068	0.022					0.375	95.0	559.0	
Bus10	0.400										0.376	94.9	559.0	
Bus11	11.000										0.768	94.6	41.1	
Bus12	0.400		0.365	0.120	0.087	0.029					0.476	95.0	702.7	
Bus13	0.400										0.476	95.0	702.7	
Bus15	20.000										6.984	90.6	201.7	
Bus16	0.400		3.800	1.249	0.859	0.282					4.904	95.0	7443.8	
Bus17	11.000										6.685	93.3	357.5	
Bus19	11.000										0.478	94.7	25.6	
Bus20	11.000										0.380	94.3	20.3	
LDMP-L2-C1	0.400		0.585	0.192	0.138	0.046					0.762	95.0	1130.2	
MV-GSB-01	11.000										5.060	92.9	270.7	
MVSB-L2-CH1	11.000										0.768	94.6	41.1	
MVSB-L4-A1	11.000										6.684	93.3	357.5	

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).
Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
						Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	MVA	%	MVA	%
Cable1	Cable	879.18	260.06	29.58					
Cable2	Cable	336.96	29.33	8.71					
Cable3	Cable	4408.80	1466.66	33.27					
Cable4	Cable	336.96	29.23	8.68					
Cable5	Cable	4408.80	1461.65	33.15					
Cable6	Cable	507.84	20.33	4.00					
Cable7	Cable	2211.84	558.96	25.27					
Cable8	Cable	507.84	25.55	5.03					
Cable9	Cable	7188.48	702.73	9.78					
Cable10	Cable	507.84	41.09	8.09					
Cable11	Cable	581.76	201.66	34.66					
Cable12	Cable	575.04	357.51	62.17					
Cable13	Cable	4079.24	1130.17	27.71					
Cable14	Cable	375.14	41.10	10.95					
Cable15	Cable	291.84	270.69	92.75					
T-Loadbank01	Transformer				5.500	5.060	92.0	4.904	89.2
TRF-HV-L1-A1	Transformer				10.000	6.984	69.8	6.685	66.9
TRF-L2-CH1	Transformer				4.200	0.768	18.3	0.764	18.2
TRF-L8-A1	Transformer				4.000	0.478	11.9	0.476	11.9
TRF-L8-A3	Transformer				1.400	0.380	27.1	0.376	26.9
TRF-L10-A1	Transformer				2.800	1.013	36.2	0.999	35.7
TRF-L10-A3	Transformer				2.000	1.016	50.8	0.997	49.9

* Indicates a branch with operating load exceeding the branch capability.

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Cable1	-8.237	-3.646	8.238	3.647	0.6	0.8	100.0	100.0	0.01
Cable10	-0.726	-0.250	0.726	0.249	0.0	-1.2	98.1	98.1	0.01
Cable11	6.331	2.951	-6.331	-2.950	0.6	0.7	100.0	100.0	0.01
Cable12	6.239	2.401	-6.238	-2.401	1.0	0.5	98.2	98.1	0.02
Cable13	0.725	0.239	-0.724	-0.238	1.5	1.4	97.5	97.3	0.24
Cable14	-0.726	-0.250	0.726	0.250	0.0	-0.3	98.1	98.1	0.00
Cable15	-4.700	-1.875	4.701	1.875	1.0	0.5	98.1	98.1	0.02
Cable2	0.953	0.353	-0.953	-0.353	0.0	0.0	100.0	100.0	0.00
Cable3	-0.940	-0.309	0.946	0.315	6.1	5.6	97.4	98.2	0.75
Cable4	0.953	0.342	-0.953	-0.342	0.0	0.0	100.0	100.0	0.00
Cable5	0.948	0.315	-0.942	-0.310	6.1	5.6	98.7	97.9	0.75
Cable6	-0.359	-0.126	0.359	0.125	0.0	-1.3	98.1	98.1	0.00
Cable7	-0.357	-0.117	0.357	0.118	0.5	0.8	96.9	97.1	0.20
Cable8	-0.453	-0.153	0.453	0.152	0.0	-1.2	98.1	98.1	0.00
Cable9	-0.452	-0.149	0.452	0.149	0.2	0.3	97.7	97.7	0.05
T-Loadbank01	-4.659	-1.531	4.700	1.875	40.4	343.4	95.1	98.1	3.01
TRF-HV-L1-A1	6.331	2.950	-6.239	-2.401	91.5	549.1	100.0	98.2	1.83
TRF-L10-A1	0.953	0.342	-0.948	-0.315	4.5	27.1	100.0	98.7	1.30
TRF-L10-A3	0.953	0.353	-0.946	-0.315	6.4	38.2	100.0	98.2	1.84
TRF-L2-CH1	-0.725	-0.239	0.726	0.250	1.3	10.9	97.5	98.1	0.60
TRF-L8-A1	-0.452	-0.149	0.453	0.153	0.5	4.4	97.7	98.1	0.39
TRF-L8-A3	-0.357	-0.118	0.359	0.126	1.3	7.9	97.1	98.1	0.98
					163.7	993.2			

* This Transmission Line includes Series Capacitor.

Alert Summary Report

	% Alert Settings	
	Critical	Marginal
Loading		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
Bus Voltage		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
Generator Excitation		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus10	Bus	Under Voltage	0.400	kV	0.389	97.1	3-Phase
Bus12	Bus	Under Voltage	0.400	kV	0.391	97.7	3-Phase
Bus13	Bus	Under Voltage	0.400	kV	0.391	97.7	3-Phase
Bus16	Bus	Under Voltage	0.400	kV	0.380	95.1	3-Phase
Bus2	Bus	Under Voltage	0.400	kV	0.390	97.4	3-Phase
Bus7	Bus	Under Voltage	0.400	kV	0.392	97.9	3-Phase
Bus8	Bus	Under Voltage	0.400	kV	0.390	97.5	3-Phase
Bus9	Bus	Under Voltage	0.400	kV	0.388	96.9	3-Phase
LDMP-L2-C1	Bus	Under Voltage	0.400	kV	0.389	97.3	3-Phase

SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	8.238	3.647	9.009	91.44 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	8.238	3.647	9.009	91.44 Lagging
Total Motor Load:	6.559	2.156	6.904	95.00 Lagging
Total Static Load:	1.515	0.498	1.595	95.00 Lagging
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	0.164	0.993		
System Mismatch:	0.000	0.000		

Number of Iterations: 3